# Tasnuba Yasmin

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## EDUCATION

**Military Institute of Science and Technology (MIST), Dhaka** **February 2019 – June2023**
Bachelor of Science**,** Biomedical Engineering **CGPA**: 3.61/4.00 (160 credit hours)

## PROFESSIONAL EXPERIENCE

**Internship Trainee** **June 2024 – Present**
Child Health Research Foundation (CHRF) Dhaka, Bangladesh

- Training on "Laboratory Techniques on Infectious Disease (LTID)" which includes:
  - Pathogen identification: Aseptic sample collection from blood and CSF
  - Antimicrobial susceptibility testing and data analysis for diagnostics
  - Specimen storage and bio-banking
- Training on "Molecular Biology" which includes:
  - DNA & RNA extraction from clinical specimen
  - Primer design for PCRs, Conventional PCR, q-RT-PCR
  - Gel electrophoresis, imaging & analysis
  - Laboratory safety protocols and waste disposal

**Research Assistant** **May 2023 – February 2024**
Cell and Tissue Engineering Lab, Biomedical Engineering Department, MIST Dhaka, Bangladesh

- **Project**: Evaluation of Mechanical Strength, Chemical Composition, and Antibacterial Activity of Nigella Sativa Extraction-Based Hydrogel.
- **Role:** Hydrogel/Agar plate preparation, Bacterial stocking and culture, antibacterial activity measurement, physical tests of hydrogel like moisture retention, water-vapor testing etc.
- **Key findings:**
  - **Mechanical Strength:** Higher Nigella Sativa conc. enhances tensile strength and rigidity.
  - **Antibacterial Activity**: Pure thymoquinone requires high concentrations for bacterial inhibition but is more effective when combined with other antibacterial compounds.
- **Skill:** Solution preparation and mixing, sterilization techniques, gelation control, physical property testing, incubation methods, preparing and using agar diffusion method (disc diffusion method), quantifying bacterial inhibition zones, analytical techniques, data analysis

**Research Intern** **October 2023 – December 2023**
Center for Health Innovation, Research and Learning (CHIRAL) Dhaka, Bangladesh
Attended workshops on:

- **"Easy Statistics for Biomedical Researchers"** covering statistical modeling, visualization, sample size determination, data analysis, and common errors.
- **Bioinformatics Bootcamp & Cancer Bioinformatics** focusing on genome sequencing (proteomics, transcriptomics, metabolomics) and database management.
- **"Fundamentals of PCR and PCR Data Analysis"** discussing different PCR types (high-fidelity, colony PCR) and analysis techniques.

**Industrial Trainee** **February 2022 – March 2022**
Combined Military Hospital (CMH) Dhaka, Bangladesh

- Learned about medical devices maintenance, repairing and hospital management.
- Received Hands on experience with biomedical devices: X-ray, CT scan, ultrasound, MRI etc.

## RESEARCH SKILLS

**Wet Lab Skills:** Antibacterial activity measurement, Bacterial Stocking & culture, Preparation of Liquid media (LB broth) and solid media (LB agar), Disc diffusion method, Zone of inhibition analysis, Nucleic acid extraction, PCR Primer design, Gel electrophoresis, imaging & analysis.
**Tools:** Magnetic hotplate, Scanning Electron Microscope (SEM), Universal Testing Machine (UTM), Laminar Flow Hood.

**Dry Lab Skills:** Biomedical Signal and Image Processing, Medical Image segmentation, 3d modeling, Computational modeling, Finite element analysis, Programming, Circuit design and simulation, Motion Analysis.
**Software:** Ansys (Fluent, Static Structural), MIMICS, SolidWorks, MATLAB, Simulink, Opensim, Kinovea, Python, Microsoft Office, Arduino.

## RESEARCH EXPERIENCE

**"Development of Garlic-Extract Incorporated PVA/Gelatin-Based Hydrogel For Biomedical Applications."(Undergraduate Thesis) April 2022 – February 2023**

**Outcome:** Garlic juice-based hydrogel had the highest antibacterial activity and endured friction like human skin. Its swelling and moisture retention capability make it ideal for wound dressings.
**Skill:** Solution formulation, gelation regulation, disk diffusion method, moisture retention assessment, swelling behavior analysis.

**"A smart and automatic wheelchair for paralyzed patient" August 2022-October 2022**

**Outcome:** Enhanced mobility and control for paraplegic and quadriplegic patients achieved through hybrid, voice, wrist, and head tilt controls, replacing traditional joystick for improved accessibility and independence. **Skill:** python

**"Design and analysis of a silicon based Toe-prosthesis" March 2022-April 2022**

**Outcome:** Created a 3D toe prosthesis model using Solidworks, validated via static structural simulation to optimize load distribution for standing, ensuring improved comfort and functionality.
**Skill:** Proficiency in 3D modeling with Solidworks and static structural simulation for load distribution analysis, design optimization for prosthetic devices.

## SCHOLARLY CONTRIBUTIONS

**Yasmin, T**, Fariha, A, Hasan, M.M (2023, May), "Development of Garlic-Extract Based Hydrogel For Biomedical Applications" –

- **Abstract Presentation**- "World Conference on Applied Science, Engineering & Technology (WCASET-23)" by IFERP (Institute for Educational Research and Publication), Malaysia.
- **Poster Presentation**-"International Conference on Pharmaceutical Sciences-2023" organized by State University of Bangladesh and the abstract of the research work was published.

## ACHIEVEMENTS

- Best paper presentation Award at "44th World Conference on Applied Science, Engineering & Technology (WCASET-23)"
- Dean's honor list (Freshman year) & Merit Scholarship (Spring '20, Fall '20)
- Government Scholarship (Talent pool) in Junior School Certificate Exam.
- "Perfect Attendance Certificate for 100% attendance" (Junior and Senior year of High school)

## EXTRA-CURRICULAR ACTIVITIES

- Obtained 2nd prize in a calligraphy competition arranged by **"MIST Literature Club".**
- Volunteered in the national event **"Anuronon-2022"** organized by "Einthoven Club" (MIST)
- Volunteer at **Blind Education and Rehabilitation Development Organization** (**BERDO**)